Find a Real Problem

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Find a Real Problem

> Prepare Find a Real Problem: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Describe a specific **user** difficulty using supplied **evidence** without inventing its cause

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a system's capability, the correctness of its **output** and permission to take an action

Reference example, not the only acceptable wording: The button establishes a capability. The fact check establishes factual support for A at that time.

The approval permits the specified classroom-board action within its limits. Public posting is outside that permission even though the button exists and A is factually checked.

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements

Reference example, not the only acceptable wording: Possible omissions include audience, source facts, required fields, format, missing-value behaviour and limits on extra claims.

C supplies **evidence**, Grade 6 specifies readers, and four labelled lines define visible fields and structure. These requirements support checks but do not guarantee compliance.

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a correct-out-of-assessed fraction and explain its denominator

Reference example, not the only acceptable wording: A has 9 correct out of 12 assessed: 9/12 = 3/4 = 75%. The denominator includes all 12 judged **outcome**s, including 3 incorrect ones.

Using 9/9 would exclude the errors and falsely report perfect assessed accuracy.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Concept to explain

A strong project begins with a specific **user** and observable **need**.

Term	Meaning
user	person affected by the problem or solution
need	observable difficulty or goal
interview	structured questions used to learn from a person
success measure	evidence showing whether the need improved

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when **need**ed.

Preparation for the connected explanation

This begins an eight-week project sequence culminating in an **evidence**-based defence. Use the activity recommender as a shared paper project; later weeks specify its catalogue and tests.

The supplied counts are fictional motivation, not collected class data or a result of an AI system. Assess problem framing separately from enthusiasm for a particular technology.

A justified decision to retain a simple method is a valid project conclusion.

Readiness answer

No. The permission covers a paper prototype and fictional requests. Collecting and publishing personal choices are different actions outside that scope.

The fictional ten-request activity **observation** in the lesson

Paper for a brief with **user**, **evidence**, **outcome**, **input**s and boundaries

The independent equipment-cupboard scenario in W4

End goal for the pictorial case

By the end, I can turn a recorded problem into a bounded, measurable project goal.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Find a Real Problem using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT observations and **interview evidence**

2. Process — the deciding step

PROCESS identify **user**, situation, **need**, and measurable effect

3. Output — the result

OUTPUT testable **problem statement** without a preselected solution

Key idea

CORE IDEA A strong project begins with a specific **user** and observable **need**.

How the mechanism works

UNDER THE HOOD Problem framing separates **evidence** about a **user**'s **need** from assumptions about a solution. A strong statement identifies who, when, what observable difficulty occurs, and how improvement could be measured.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE 'Students **need** AI' is not a **problem statement**; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

Step 1

STEP 1 — OBSERVE **observation**s and **interview evidence**

Step 2

STEP 2 — TRACE identify **user**, situation, **need**, and measurable effect

Step 3

STEP 3 — CONCLUDE testable **problem statement** without a preselected solution

Model answer

MODEL REASONING A strong statement names the **user**, context, observable difficulty, and desired improvement without requiring AI. Protect privacy, avoid leading questions, and distinguish what the **user** said from your inference.

A claim to avoid

DO NOT OVERCLAIM Do not collect personal stories unrelated to the design question.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

Shared markers are often lost because tray labels are unclear.

The teaching case

State the **need** before selecting AI.

Exact explanation to model

Why the result follows

Students **need** to return markers correctly and quickly. Test clearer labels before proposing a model.

Explain the added worked example

Explain the added worked example · 1

For the final project sequence, we will develop a small activity recommender on paper. Its **user**s are fictional pupils choosing a classroom activity that fits their available time and room conditions.

We will decide what the recommender must do, build a traceable **process**, test it and explain its limits. A recommendation is a suggestion; the pupil can decline it.

Explain the added worked example · 2

Start before the solution. Our supplied **observation** log describes ten fictional requests made using an activity catalogue. Six first choices fit the request's time and room conditions. Four did not and **need**ed another choice.

These invented classroom records give us a problem to investigate. They are not **observation**s of your classmates and do not show how often this happens in other classes.

Explain the added worked example · 3

A useful **problem statement** is: “In the supplied ten-request exercise, four first choices did not fit the stated time or room conditions, so those **user**s had to choose again.” “We **need** AI” names a possible technology but says neither who **need**s help nor what should improve.

The log also does not establish whether unclear wording, missing information or another cause produced the mismatches.

Explain the added worked example · 4

Our target **outcome** is a compatible suggestion when the catalogue contains one, and a clear no-match message when none fits.

Compatibility means respecting the stated conditions; it does not mean that every pupil will enjoy the activity. We can check time and room conditions from a request and catalogue. Enjoyment would **need** different **evidence**.

Explain the added worked example · 5

Count compatible suggestions, incompatible suggestions and no-match messages separately. Otherwise a system could avoid every wrong suggestion by refusing to suggest anything.

If a valid option existed, that refusal would still fail to provide the intended help. Record the total number of requests so a count has a clear denominator.

Key terms

Term	Meaning
User need	Something a person needs to accomplish in a particular situation
Problem statement	A specific description of who faces a difficulty and what that difficulty is
Outcome	The result experienced by the user
Constraint	A condition that a proposed solution must respect
Success measure	An observable quantity or result used to judge progress

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Project choice	Boundary for this sequence	Reason
Users and setting	Fictional classroom activity requests	A clear situation to test
Information	Available minutes and quiet-room requirement	Relevant to compatibility
Action	Suggest an activity or report no match	The user retains the choice
Outside the project	Names, profiles, purchases and public posting	Not needed for this paper task

Model the reasoning aloud · 6

The teacher permits a paper prototype using fictional requests and an approved catalogue. We do not **need** names, home details or a history of a pupil's choices.

A request can say "10 minutes, quiet room required" without identifying anyone. Use only the request information **need**ed to test the intended **outcome**.

Model the reasoning aloud · 7

A clear catalogue or a short checklist may solve this problem. We will compare a proposed recommender with a simple starting method instead of assuming a more complicated system is better.

This project can use explicit rules; it does not have to fit a machine-learning model. The choice should follow the problem, available **evidence** and limits of the task.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move	Teacher prompt
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Find a Real Problem

A fictional class loses time finding the correct craft tray.

The facts we will use

Supplied facts

In eight recorded requests, three selected trays lack a required tool; five match.

Supplied facts

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

AI ACADEMY / AI-1/29

Find a Real Problem

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Observe problem

Three mismatches are recorded out of eight requests.



2. Define need

Help users choose a tray containing every requested tool.



3. Set success test

Compare selected tray against the approved inventory for each new case.



4. Bound the project

Use fictional tool requests; do not collect identity or purchase equipment.

CHANGE ONE THING

The approved list contains no tray with every requested tool.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Observe problem	Three mismatches are recorded out of eight requests.
2. Define need	Help users choose a tray containing every requested tool.
3. Set success test	Compare selected tray against the approved inventory for each new case.
4. Bound the project	Use fictional tool requests; do not collect identity or purchase equipment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

What this example does not establish

Keep the stated limit

Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

Change one thing in the pictured case

Change one thing

Change: The approved list contains no tray with every requested tool.

Predicted result

Prediction: The finder should report no suitable tray rather than guess.

Why the result follows

Reason: Success includes honestly identifying unsatisfied requirements.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Find a Real Problem, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Shared markers are often lost because tray labels are unclear.

2. State the decision to explain

Write the question to explain: State the **need** before selecting AI.

3. Reveal the worked reasoning

Reveal after **prediction**s: Students **need** to return markers correctly and quickly. Test clearer labels before proposing a model.

Hold back the independent task

Keep for an independent attempt: Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The brief should identify fictional pupils choosing classroom activities, four incompatible first choices out of ten, and the goal of a compatible suggestion where one exists or an explained no-match result otherwise.

Inputs are minutes and quiet-room requirement. Boundaries include fictional requests, teacher-approved catalogue and suggestions only. A clear catalogue or checklist is a simpler alternative.

The log does not establish causes, feelings or a future tool’s benefit. Exit: no; always asking avoids a wrong suggestion but may miss available compatible activities.

Check useful matches and justified no-match cases, not only avoided errors.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the limits of permission for a paper prototype.
7 to 20 minutes	Read the observation and distinguish difficulty, cause and proposed solution. Complete W1.
20 to 33 minutes	Trace the evidence-to-outcome diagram and complete W2.

Stage	Teacher action
33 to 45 minutes	Select relevant inputs and a simple alternative in W3.
Session 2 start 0 to 8 minutes	Retrieve why four mismatches do not prove a cause or a solution.
8 to 23 minutes	Draft the guided project brief; challenge unsupported claims and excess data.
23 to 38 minutes	Collect W4 independently, checking the brief against the new scenario.
38 to 45 minutes	Complete W5 and the exit; discuss how a comparison could favour the simple method.

Reduce help deliberately

Completion practice

Completion task: A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

Reasoning to listen for

Teacher check: The log establishes four incompatible first choices out of ten supplied fictional requests.

It does not establish every pupil's opinion, the cause of the mismatches or that an untested recommender will improve **outcome**s. Accept any two of those unsupported conclusions with the bounded **observation**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If **need**ed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for A problem worth solving. Proposed explanation: “Using AI is itself the **outcome**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

> #### Expected repair
>
> Expected correction: State what improves for the **user** and how to observe it.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Write a one-sentence project goal and say what **evidence** would count as a failure.

Reasoned teacher answer

Goal: choose an inventory-confirmed tray containing all requested tools or report none. A selected tray missing a required tool is a failure.

Changed-case reasoning: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Find a Real Problem investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: **Interview** a classmate and write one testable **need** statement.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR Every classroom problem **need**s AI.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Using AI is itself the **outcome**: State what improves for the **user** and how to observe it.

A mismatch count proves its cause: Separate **observation**s from explanations that **need** further **evidence**.

No wrong suggestions means success: Check whether valid options were missed or every request was refused.

A prototype requires real personal data: This paper project uses permitted fictional requests and task-relevant **input**s.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Find a Real Problem questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Students **need** to return markers correctly and quickly. Test clearer labels before proposing a model.

Independent question

Independent question · withhold the key

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

Expected independent answer

Teacher key · Expected independent answer

The problem is finding who borrowed a book and where it belongs. Compare a labelled shelf and a checkout record with an AI approach. The solution should address the actual **need**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Every classroom problem **need**s AI.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Time students finding predefined books using each approach, record wrong locations and assistance **need**ed, and use comparable book sets and conditions. Start with the simpler label trial.

Choose based on **evidence**, effort and privacy **need**s, not the presence of AI.

Explanations for the additional workbook examples

W1 Write the problem without prescribing a tool

Workbook question · W1

Use the ten-request activity log to write a **problem statement** with a **user**, setting and observed difficulty. Explain why “Build an AI app” is not yet a **problem statement**.

Teacher reasoning · W1

For example: fictional pupils choosing activities in the supplied classroom exercise had to choose again in four of ten requests because the first choice did not fit the time or room conditions.

Build an AI app names a solution or construction task without specifying a **user** difficulty or **evidence** of **need**.

W2 Identify claims beyond the evidence

Workbook question · W2

A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

Teacher reasoning · W2

The log establishes four incompatible first choices out of ten supplied fictional requests. It does not establish every pupil's opinion, the cause of the mismatches or that an untested recommender will improve **outcome**s.

Accept any two of those unsupported conclusions with the bounded **observation**.

W3 Choose information that serves the outcome

Workbook question · W3

The paper project must suggest an activity compatible with available time and a quiet-room requirement. Choose relevant **input**s from minutes available, quiet required, pupil name and home address.

State the **output** and a simpler alternative to a recommender. Explain one boundary.

Teacher reasoning · W3

Minutes available and quiet required are relevant. Name and address are unnecessary. **Output** a compatible activity suggestion or an explained no-match result.

A clearer catalogue or compatibility checklist is a simple alternative. Keep requests fictional and the action advisory; do not extend the task to collecting personal profiles or publishing choices.

W4 Frame a different problem independently

Workbook question · W4

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

Teacher reasoning · W4

Teams selecting equipment in the supplied exercise had three mismatches out of eight and **need** a box whose listed contents meet the request, or a clear no-match result.

Use requested items and the inventory; personal identities are unnecessary. Exclude real purchases, changing stock or public posting. A clearer inventory or item checklist is a simple alternative.

The counts do not explain why teams chose mismatching boxes or prove that a box-finder will improve selection.

W5 Measure help as well as avoided errors

Workbook question · W5

For W4, a proposed box-finder never selects a wrong box because it always reports no match. Explain why counting only wrong suggestions would hide a failure.

Name two results to record on test requests and explain why the original three-of-eight **observation** is not already an evaluation of the new tool.

Teacher reasoning · W5

Always refusing misses useful matches when suitable boxes exist. Record compatible or incompatible suggestions and whether no-match **output**s were justified by the inventory, along with total requests.

The three-of-eight **observation** describes the earlier selection **process**; the proposed tool has not yet been tested under a comparison procedure.

Assess independent transfer

Independent question · withhold the key

Selected case: In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

Selected case answer

Teacher key · Selected case answer

Teams selecting equipment in the supplied exercise had three mismatches out of eight and **need** a box whose listed contents meet the request, or a clear no-match result.

Use requested items and the inventory; personal identities are unnecessary. Exclude real purchases, changing stock or public posting. A clearer inventory or item checklist is a simple alternative.

The counts do not explain why teams chose mismatching boxes or prove that a box-finder will improve selection.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

Connect Find a Real Problem to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Choose the Classroom Sorter's real **user** and measurable **need**; keep an AI-free solution option open.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 30: Choose **Input**s and **Output**s. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use sentence starters: In this setting, these **user**s **need**; the supplied **evidence** shows; we will check; we will not change. Allow an oral brief recorded by the teacher without supplying the choices.

Extension: describe **evidence** that would make the team abandon its preferred solution.

W4 must transfer the structure: three of eight mismatches, a compatible box or justified no-match **outcome**, requested items and inventory, one boundary, a simple alternative and a bounded claim.

After support use a new scenario: six fictional room requests, two assigned rooms too small, four suitable; permit paper recommendations only.

Require capacity and group size as relevant **input**s, no real bookings, and no invented explanation for the two failures.

Week 30 turns the activity problem into explicit requirements and a simple baseline for comparison.

Plan the delayed return

Suggested return points in this level: ai-1/30, ai-1/32, ai-1/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Find a Real Problem lesson; it is a

checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Describe a specific **user** difficulty using supplied **evidence** without inventing its cause

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem.

Identify two unsupported conclusions and state what the log actually establishes.

Supported practice answer

Reasoned practice answer: The log establishes four incompatible first choices out of ten supplied fictional requests.

It does not establish every pupil's opinion, the cause of the mismatches or that an untested recommender will improve **outcome**s. Accept any two of those unsupported conclusions with the bounded **observation**.

Check every learner before discussion: Use the worked case for A problem worth solving. Proposed explanation: "Using AI is itself the **outcome**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

Repair the misconception

Misconception repair: State what improves for the **user** and how to observe it.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The approved list contains no tray with every requested tool. Require a **prediction** and a reason before comparing. Expected result: The finder should report no suitable tray rather than guess.

Reason: Success includes honestly identifying unsatisfied requirements.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

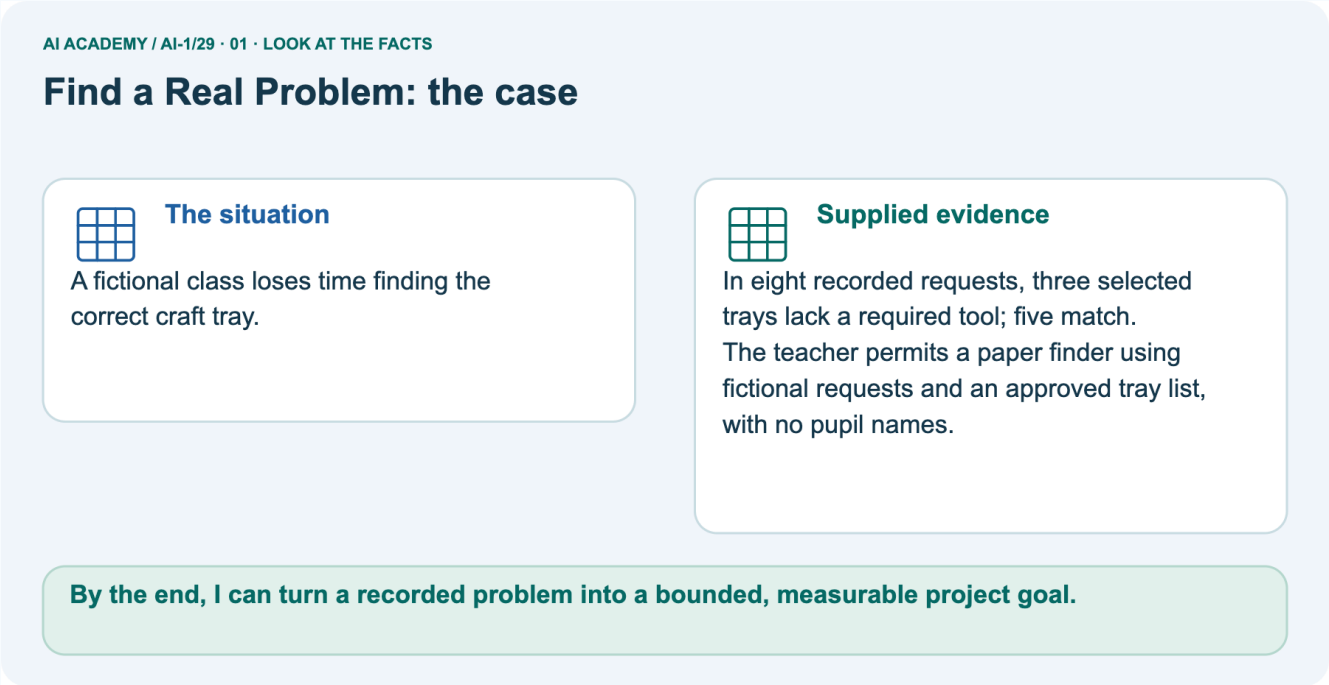
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Find a Real Problem graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Find a Real Problem: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional class loses time finding the correct craft tray. Supplied facts: In eight recorded requests, three selected trays lack a required tool; five match.

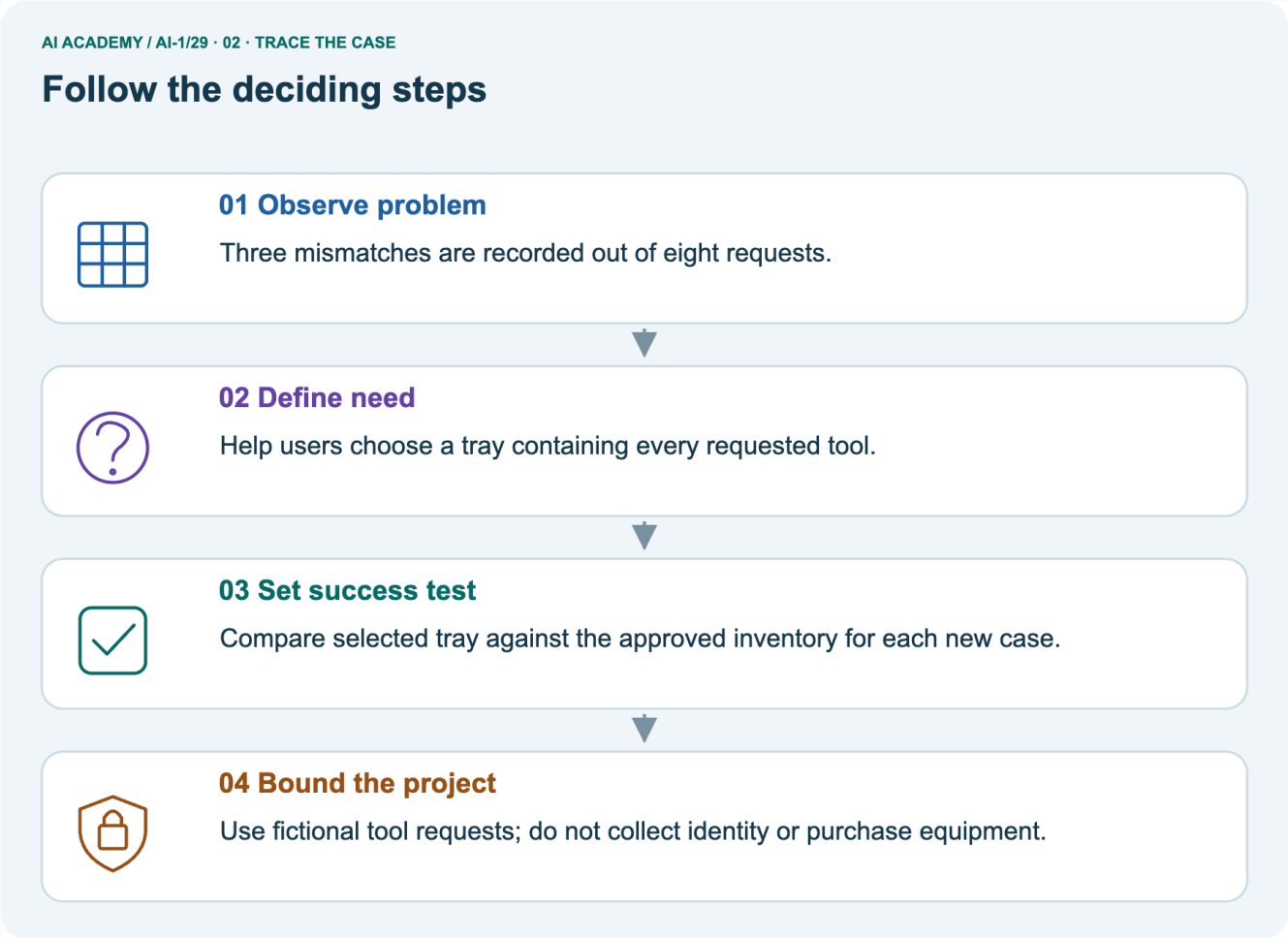
The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

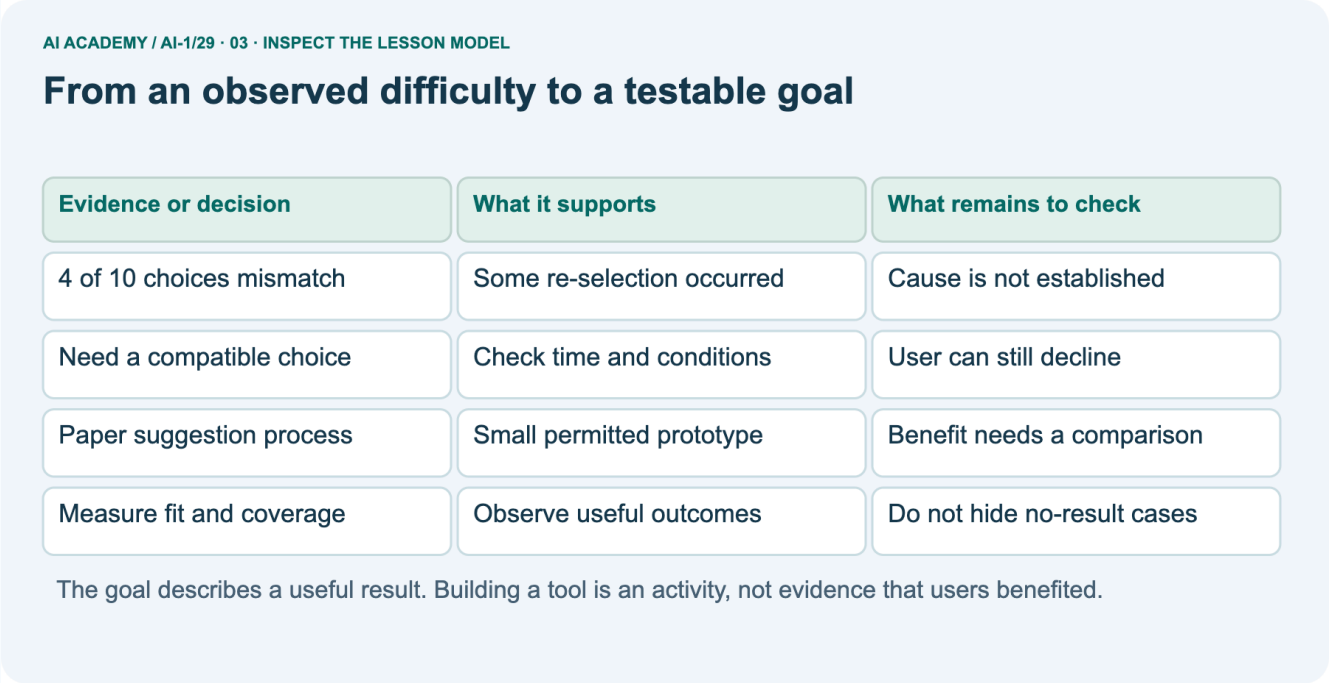
Observe problem: Three mismatches are recorded out of eight requests. → Define **need**: Help **user**s choose a tray containing every requested tool. → Set success test: Compare selected tray against the approved inventory for each new case. → Bound the project: Use fictional tool requests; do not collect identity or purchase equipment.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. From an observed difficulty to a testable goal



Follow the labelled picture with the matching explanation. Open the vector diagram

From an observed difficulty to a testable goal. 4 of 10 choices mismatch | Some re-selection occurred | Cause is not established. **Need** a compatible choice | Check time and conditions | **User** can still decline.

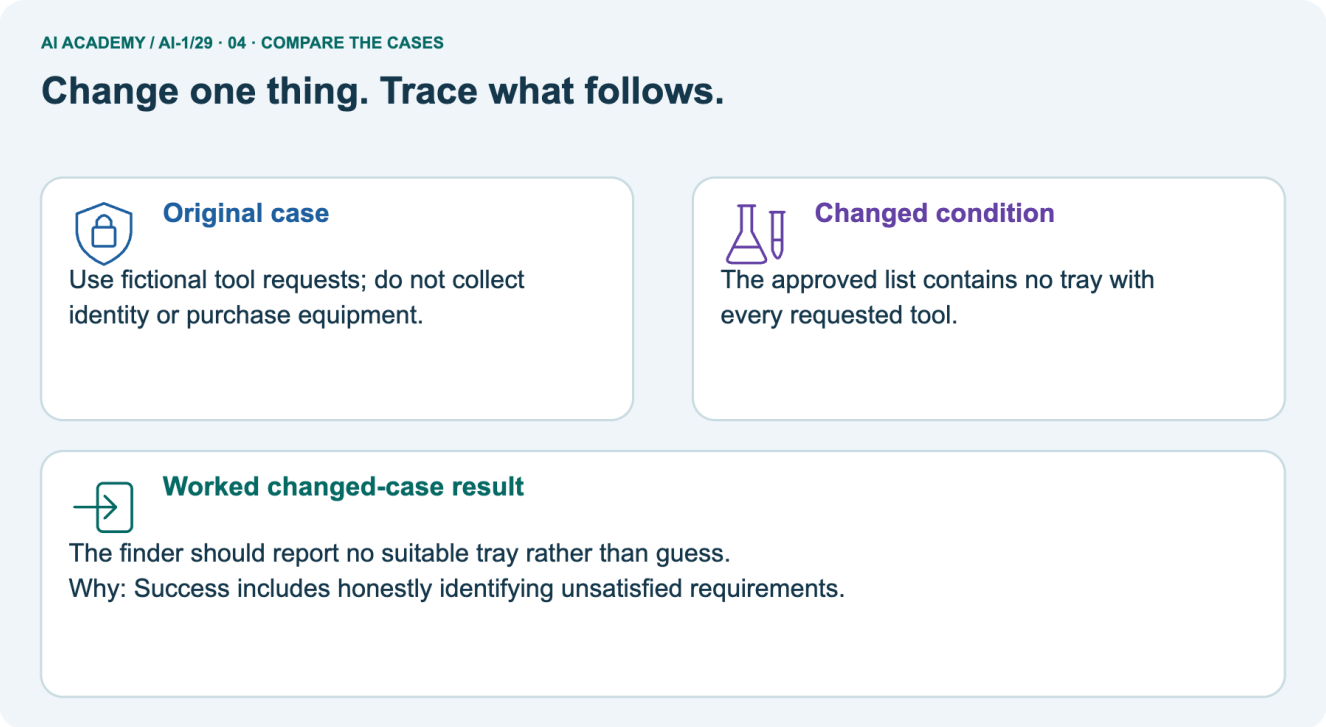
Paper suggestion **process** | Small permitted prototype | Benefit **need**s a comparison. Measure fit and coverage | Observe useful **outcome**s | Do not hide no-result cases

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use fictional tool requests; do not collect identity or purchase equipment. Changed condition: The approved list contains no tray with every requested tool.

Expected result: The finder should report no suitable tray rather than guess. Reason: Success includes honestly identifying unsatisfied requirements.

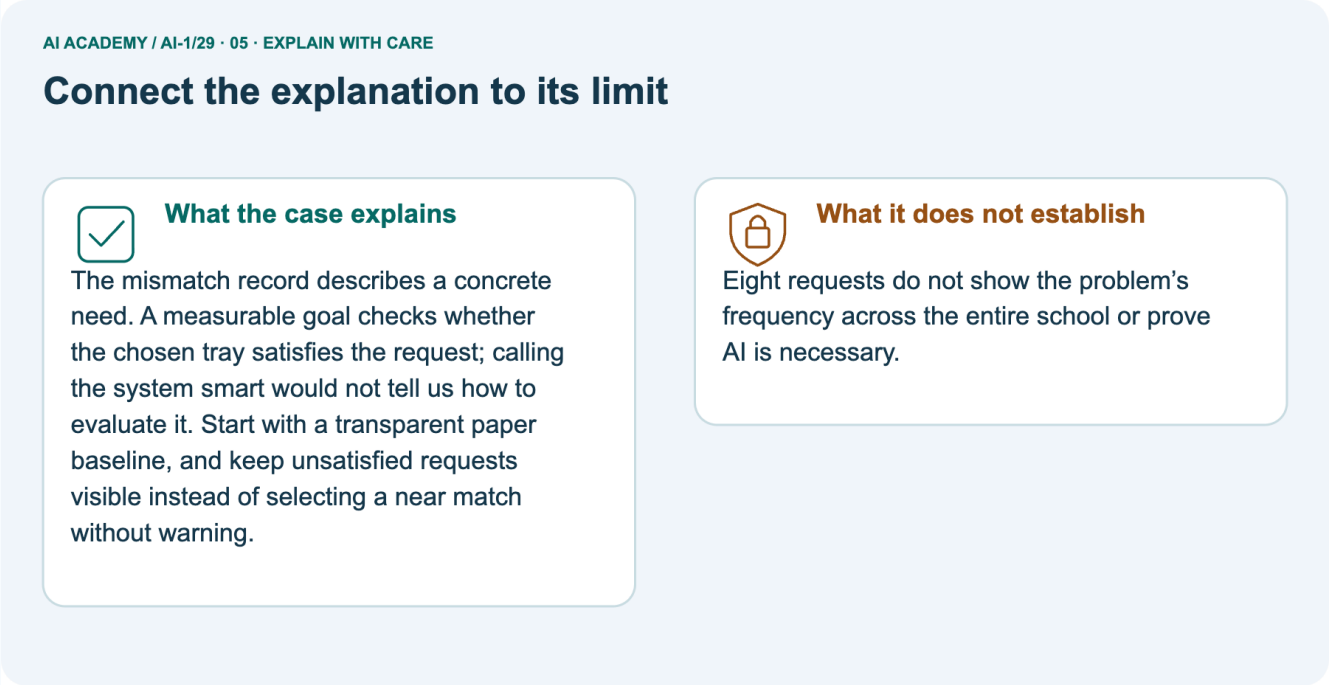
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Success includes honestly identifying unsatisfied requirements.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Limit: Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

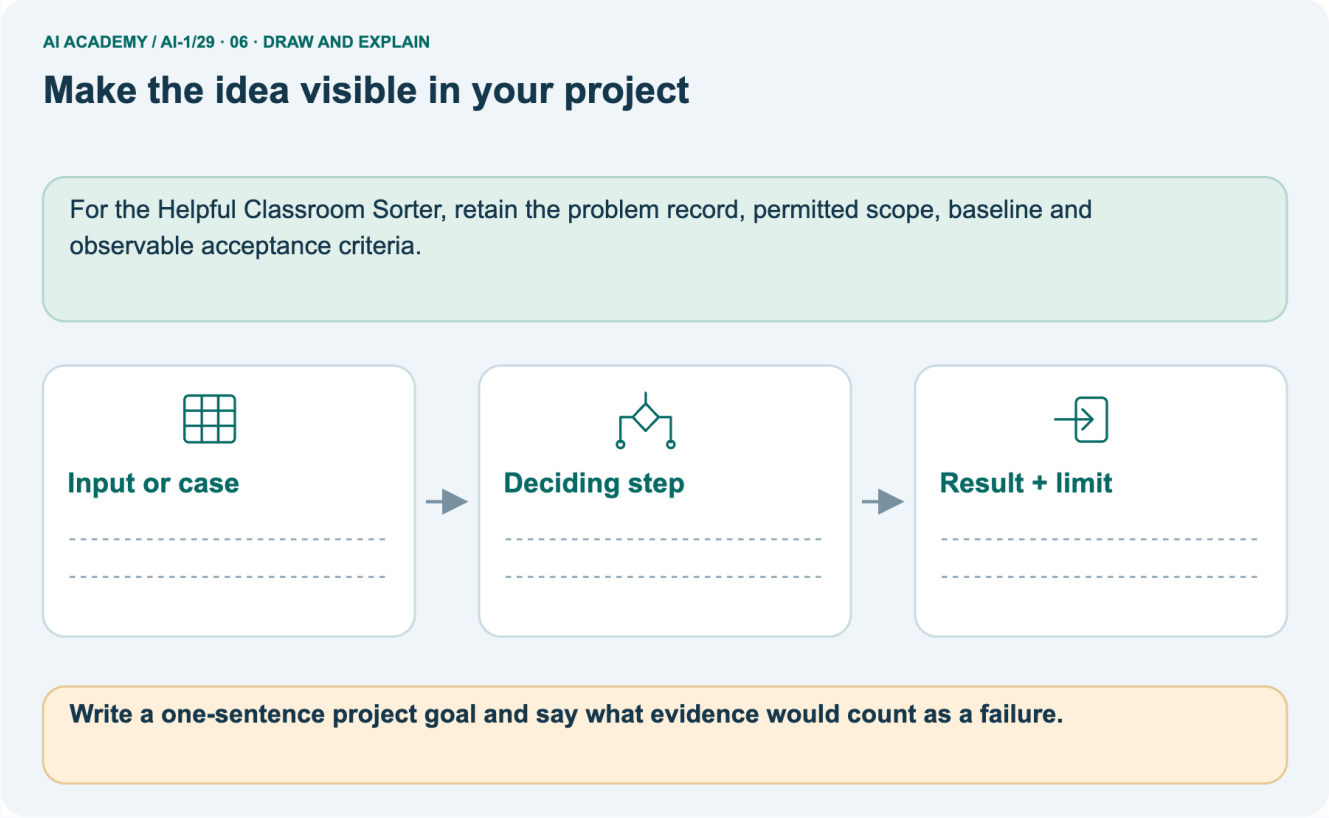
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria. Draw your **input**, deciding step, result and limit.

Supported question: Write a one-sentence project goal and say what **evidence** would count as a failure.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Goal: choose an inventory-confirmed tray containing all requested tools or report none. A selected tray missing a required tool is a failure.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.